

Fundamentals of Information Systems Security

Lab 2: Network Reconnaissance with Nmap

Objective: Perform basic network scanning and service enumeration.

Prerequisites

- A running Kali Linux VM (from Lab 1).
- Network connectivity (NAT or Bridged).
- Target machine (can be localhost, another VM like Metasploitable 2, or a lab partner's machine with permission).

Learning Objectives

By the end of this lab, you will be able to:

- Identify your network subnet and IP addressing.
- Perform host discovery (ping sweeps).
- Conduct TCP/UDP port scans.
- Determine operating systems and service versions.
- Use NSE (Nmap Scripting Engine) for basic vulnerability detection.

Lab Steps

Step 1: Determine Your Network Environment

Open a terminal and run:

```
ip a
```

Identify your IP address and network mask (e.g., 192.168.1.10/24). Your network range is 192.168.1.0/24.

Step 2: Ping Sweep (Host Discovery)

Perform a ping scan to find live hosts on the network:

```
nmap -sn 192.168.1.0/24
```

Note: Replace 192.168.1.0/24 with your actual subnet.

Step 3: Scan Localhost

Scan your own machine to see open ports:

```
nmap 127.0.0.1
```

Step 4: Stealth SYN Scan and Version Detection

Choose a target host (e.g., 192.168.1.1 for the router, or a lab VM). Run:

```
nmap -sS -sV -O 192.168.1.1
```

Breakdown of flags:

- **-sS**: SYN stealth scan (requires root privileges).
- **-sV**: Probe open ports to determine service/version info.
- **-O**: Enable OS fingerprinting.

Step 5: Port Range Scan

Scan all 65,535 ports on a specific target:

```
nmap -p- 192.168.1.1
```

Warning: This takes a long time. Consider scanning a specific range like -p 1-1000 for testing.

Step 6: UDP Scan

Scan for common UDP services (DNS, SNMP, DHCP):

```
nmap -sU -p 53,67,68,161 192.168.1.1
```

Step 7: Nmap Scripting Engine (NSE)

Use default scripts to detect common vulnerabilities:

```
nmap --script=default 192.168.1.1
```

To scan for specific vulnerabilities (e.g., SMB):

```
nmap --script=smb-vuln* 192.168.1.1
```

Deliverables

Submit a report containing:

1. A screenshot of your **ip a** output.
2. A screenshot of the ping sweep results.
3. A screenshot of the **-sV -O** scan output for your target.
4. A table listing the target's open ports, services, and OS guess.
5. A brief explanation of why SYN scans are considered "stealthy."